Fake Social Media Profile Detection and Reporting

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***Abstract*— The increasing amount of fake accounts on social media poses significant threats to the authenticity of digital interactions and amplifies the risk of threats related to false information, fraud, and digital abuse. Although advanced fraud detection systems regulation have been designed and implemented in detection systems in both the private and public sectors, these alternate systems are, in general, not usable (or available) to the average person because of a combination of intricacy and a disparity of significant resource expenditures required to run these systems. This research endeavours to fulfill these needs by offering a practical, open, and adaptable fake profile detection system in social media. The identified system is based on simple rule-based systems so that it is interpretable and simple to use. The system permits users to provide URLs to suspected fake accounts, and the system retrieves public data from the social media site from which the user provided the URL, calculates a number of risk indicators: ratio of followers to those whom the user is following, completion percentage of the profile, levels of entropy in the user name, and degree of similarity in the user-generated content, and produces a risk level based on these indicators, all the while generating a risk score for the profile and providing a detailed report that is simple and readable. The system provides extensive reporting capabilities to the users, offering a comprehensive report to the users even allowing users to edit reports in the system, and it ensures that the users are provided with additional data and growing processes at the their convenience, ultimately resulting in the users in being better positioned to make finely differential moderated actions. The system is open-source and built.**

Keywords: Fake Profiles, Social Media Security, Account Detection, Rule-Based Analysis, Django, Web Application, Cybersecurity.

I. INTRODUCTION

The identity of the virtual and real worlds coupled with the role of social media such as Facebook and Twitter have transformed radically and it incorporates social media as a communicating tool, expressing oneself through identity and individuality, and interconnection and connections globally. Nonetheless, the globalized world has an advanced network of artificial and in many cases deceitful accounts. These automated accounts and fake identity accounts have some negative consequences such as identity theft, financial asset fraud, misinformation, social manipulation, and depression control and manipulation.

These fraudulent accounts also operate on a large scale, eroding client trust, and undermining civil discourse while exacerbating digital and general skepticism and insecurity. These fraudulent accounts have also been a threat to all the online social platforms. To these fraudulent accounts, online social platforms have implemented proprietary algorithms that are deep learning and behavior analytics based to try to identify and remove these accounts. These platforms are losing millions of accounts each year to these algorithms and accounts daily. These systems are primarily reactive algorithms that lack transparency, and struggle to outpace the learning systems of digital profile manipulators. Experts have proposed multiple account fake detection systems integrating machine learning that have verified experimental accuracies that exceed 95%. These systems, however, have exorbitant computational requirements, necessary extensive labeled datasets, and lack real interpretability, and thus, are not meant to be integrated by individual small enterprises lacking the relevant computational resources.

Users can submit profile URLs for analysis through a web interface built for this purpose, while the admin dashboard will provide the centralized profile management, report generation, and analytics monitoring. The framework is designed to facilitate the users' engagement with each other, and assist them in safely navigating the Internet while maintaining accountability, transparency, and a community-driven approach. Technologies like Django and PostgreSQL demonstrate that the research can integrate social inclusivity with strong cybersecurity, digitally restoring trust in communication and demonstrating that, technically speaking, strong cybersecurity is possible in today's ecosystem.

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II. Literature Review

As concern grows over the growing number of automated and fake accounts, research on countermeasures is rapidly expanding. Ferrara et al. [1] bot activity on Twitter has analyzed what is known about detection and more sophisticated classifiers, the scalability of imbalanced data, and the need for detection systems to adapt to more complex countermeasures. Dash et al. [2] contributed to the field of bot detection systems focusing on integration of text and contextual activities through the hybrid CNN-LSTM model. Uppoor et al. [3] of the field of bot detection systems applied to complex relational account structures of inauthentic behavior to advanced Graph Neural Networks (GNNs); a new approach to learning representation in graphs. Wu et al. [4] detection model MULTI which augments classification by integrating text, imagery and metadata in a unified multi-modal framework. Continuing in the same direction, Ng et al.

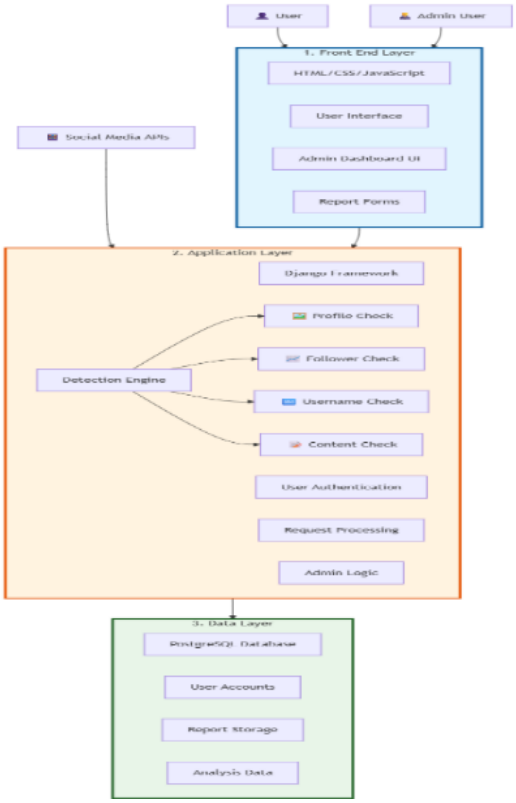
Al-Rakhami et al. [5] presented DeepBot employing deep learning methods with sophistication and with temporal and content attributes to identify social bots in multiple platforms. Al-Rakhami et al. [6] acknowledged the need to adjust to the domain for real-time deep learning for the detection of spam and fraudulent accounts on Instagram. Cresci et al. [7] on the other hand, elaborated on the history of the detection of static bots on the basis of deep heuristics "DNA" to the adaptive and real-time "RNA" bots and models, describing the perpetual arms race therein. For adaptive, dynamic behaviors, Zhang et al. [8] proposes the first of its kind, transformer-based framework for the social bots detection, for it surpassed other recurrent methods. FakeNet, a deep neural network for cross-platform detection of fraudulent accounts, was developed by Sharma et al. [9]. Only Nguyen et al. [10] addressed the privacy issue by the decentralized detection of data kept within the users device employing federated learning.

III. PROPOSED METHODOLOGY

The suggested multi-modal framework for spotting fraudulent social media accounts was refined during a methodical and progressive engineering procedure which focused on merging cyber security theory and practice through user centered design. The engineering design uses a theory of engineering and software design which organizes activities during the engineering cycle. The framework is engineering and software based and performs critical social and technical adjustments at essential points in an engineering software life cycle including requirements collection, modular design and architecture, stack or platform engineering, and multi-phase systems evaluations. Proximity of the models and their ability to streamline social, economic and technical security to deliver feasible solutions was critical. The influence of academic work on social bots and systems in design, and the systems usability in conjunction with the social bots were also iterative to design and the architecture of the work. The ultimate objective is to restore some form of design which allows the system to circumvent the more complex systems in use for computational work, and to provide a high degree of usability in an adaptable framework for a web based detection system.

Explainability is one of the core principles of this framework. In avoiding complex 'black box' machine-learning systems, this framework emphasizes transparency and provides users the means to understand the process of how and why a particular risk classification is made. This fosters trust and responsibility, and increases users' active participation in the process of detection. In other words, this framework is the result of a blend of theoretical scholarship and hands-on engineering, the purpose of which is to produce a scalable, open, and democratized model for the active participation of both technical administrators and lay users in protecting social media ecosystems.

A. System Architectural Diagram



1.1 System Architectural Diagram

1. Presentation Layer: It is an interface developed using html5, css3 and javaScript which is reponsible for providing web responsive experience on any device or screen sizes. The

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UI is easy to get up and running with, leading the user through interaction by analytical response on interaction conditions as well as output visualizations.

2. Application Layer: Application around the Django web framework because of its strength, modular MVT (Model-View-Template) design and security concern not to developing a web application on a framework with interconnectivity with social medias. In this layer, there are the rule-based detection engine, user authentication, user queries and real-time querying of social media profile data from external platforms (i.e., Twitter and Facebook) using APIs. Token-based authentication and request validation in the application layer is also used to stop unauthorized user access and attack surface.

3. Data Layer: This layer Here the basic system data (e.g., user data, report submission data, analytical result data, administrative data etc.) are securely stored. The ORM (Object Relational Mapper) of Django is utilised to model the data layer that interfaces safely and responsibility with application layer for generating output in a formatted bound format.

The three layers combined provide for a solid and sustainable architecture. Each layer is autonomous but they are integrative as a whole to enable unlimited scalability and components that can be easily upgraded in the long run in more than one technical environment.

B. Rule-Based Detection Engine

At the core of our suggested architecture is a rule-based detection system. It uses common sense heuristics derived from empirical evidence and behavioural science. This contrasts with black-box and highly complicated machine-learning models that offer no interpretability of the decisions while this method aims at maximum transparency and interpretability. A user or admin can quickly understand why an alert fired by only understanding the rules in the detection engine.

Profile Completion Analysis: This section analyzes whether people have a bio, a profile picture, and a location. The profiles lacking these details are more fraud risks as a more thoughtful profile (as long as genuine and not by bots or automation) risks less digital fraud.

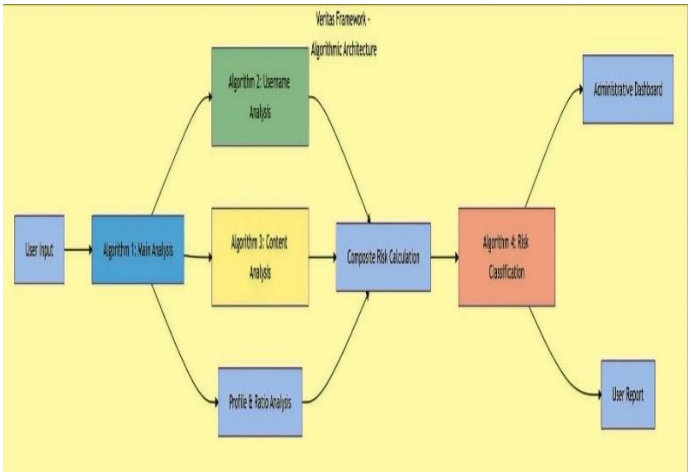
Follower and Following Analysis: In this module, the ratio between a user's followers and followed users is calculated. An uneven follow ratio is found, and the example of lopsided ratio are usually related to purchases of followers or not follow bots (which for these users a deeper analysis is more common).

- **Usernames Entropy Analysis:** This detects nonsensical random Login IDs which are usually linked to automated account registration by bots. Real usernames have lower, entropy and the more substantial ones are surely for automated usernames.
- **Analysis of Content Similarity:** This section recognizes the presence of Duplicate and near-duplicate content in COS user posts through cosine similarity. A telltale sign of posting too much content is certainly the presence of a bot or the account having mass/multiple POST authority.
- **Posting Frequency Analysis:** This part of the system evaluates the time gap between weekly posts, and their regularity. Accounts that tweet on the hour, every hour, or at the same time every day, or week, are identified as potentially automatic and characteristic of non-human behavior.

All the heuristics provide weighted average to a composite risk score that is subsequently normalized into one of the three risk categories: Low Risk, Moderate Risk or High Risk. The system produces an analysis report which is readable by people and provides a description of the reasoning that the classification is based upon and why the classification made. This guarantees that in addition to being offered actionable recommendation to the users, recommendations are also given in a transparent manner on how the results were identified.

C. Algorithms

The structures of the framework help keep everything standardized, thus helping in keeping everything uniform. Each of these units represents the core of one computational process.



1.2 System Integration Diagram

Algorithm 1 Profile Evaluation Flow: Consolidated controller from data collection and preprocessing to rule evaluation and result collection.

Algorithm 2: Entropy of Username Calculation This section calculates the entropy and detects abnormal statistically usernames.

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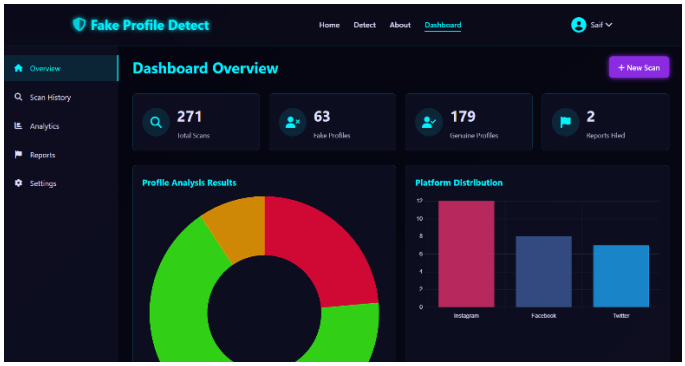
Algorithm 3: Detection of Redundant Content – Textual posts are compared for interspersion e.g., through like cosine similarity or an index of Jaccard.

Algorithm 4: risk classification and pred admin discretion: Aggregated heuristics results enable composite. pred if we preserve min. queries. scores, creates the respective reward classes and signals the required admin actions.

Each of the algorithms has a modular design and could be elongated as the system scales or could be augmented/combined with ML-based modules/adaptive weights while keeping the core logic intact.

1.3 User Interface



1.4 User Dashboard Interface

D. User and Dashboard Interfaces

As we have forecast, the Dual-Interface provides hassle-free accessibility and full control. In respect to the User Dashboard, front-end users can report, view status, and summary analysis of reports. The reports were visually anchored and presented in a narrative manner to explain the purpose of the report to ensure transparency and to engage communities. In respect to the Administrator Console, an exclusive tier of access with full visibility of all system analytics, risk tier, and history of system data, wherein admins can interact with automated category suggestions, exercise report judgment, and review analytics data of the system. This will bring a balance between automated detection and human supervision.

IV. EXISTING SYSTEMS AND THEIR LIMITATIONS

The active and sophisticated task of fake social media account detection is a constant engagement. Misconduct learn themselves, and platforms and researchers focus their attention to reactive defense measures. In the past decade, a plethora of fake account detection has been developed in both industry and academia. However, very few are truly comprehensive and accessible in a practical sense by the average end user.

In general, there are three categories of current detection methodologies: proprietary framework systems, manual reporting systems, and AI-driven models made in university or market contexts. Each of these aids in account fraud, but there are also significant challenges which include: opacity, lag-time, flexibility, scalability, and user-driven agency. Hence, there remains a disparity between ease of practical application and public trust.

A. Proprietary Platform Systems

Meta, X (formerly Twitter), and TikTok, all major players in the social media field, deploy proprietary, real time, automated fraud account detection systems leveraging neural networks, natural language processing, and behavioral analytics. These systems, which fraudulently operate millions, while effective and automated, are not transparent and lack any meaningful public accountability. Lack of follow-up leads users to distrust a system designed to operate without user input. There are no fraud detection systems where users operate the functionality to stop or mitigate the risk to their accounts. Custom lack of transparency functions to provide systems with no public accountability.

B. Manual Reporting Mechanisms

Reporting fake accounts manually is still among the most widespread methods of detection. It gives users some ability to help manage the content. But it is a slow, horizontal, opaque system. Most reports receive no response, causing user frustration and a loss of motivation. Given the number of accounts that are active on most modern platforms, there is no way that manual moderation can keep up. And so, bad accounts remain active for long periods of time, counterfactually engaged, and spreading misinformation and other fraudulent content.

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C. AI Models in Acadamia and Industry

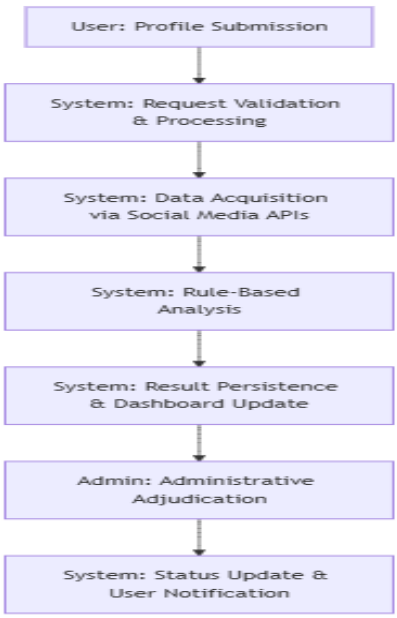
In the recent past, academia and industry-based research has been conducted on the development of AI-based fake profile-detecting models, in spite of being sufficiently good in academic and business uses, such models are at less likely to reveal a real-life solution to actual use, and, hence, the low scalability factor.

D. The Technological and Social Gap

Detection Technologies continues to progress; however, they remain largely unavailable to the layperson, especially in usability. Proprietary systems can be powerful, The new framework has the purpose of closing this gap through the layering of heuristics from validated academic work, molding them into a rule-based, web-deployable system that is heavily centered on interpretability and user engagement. The framework aims to actionable detection that is also transparent and community-driven.

V. SYSTEM WORKFLOW

The way the system will work including the engagement of the user and the smooth feedback is properly structured into phases. High precision of the algorithm and human discernment create huge accuracy, equality, and satisfaction to the users in each phase of the process. This will occur in the following manner:



2.1 Work Flow Diagram

1. Profile Submission: The first step in creating a suspicious social media profile is a registered user submitting the profile in a user-friendly web interface. A URL or profile ID is entered by users, and all further analyses are drawn up. This step shows that there is community participation of establishing trust and integrity within the social media sphere.

2. User Request Compliance: The system performs thorough checks on a user's security profile for every request. The system starts by checking user session credentials first to ensure there is no unauthorized access into the platform. The system then checks if the submitted URL or identifier is properly formed and if the URL or identifier correlates to the targeted platforms. The system adds domain of the request to the exception list to perform originating request checks. This is to prevent attacks and unauthorized data entry through different forms of the domain. The system moves to the next stage only if there are no checks failed.

3. Accessing APIs: The system is now ready to access the external APIs. The remaining access points are the Twitter API and the Facebook Graph API. The API is processed to public metadata and secure information processed by the system and the information is kept in the database. The database keeps the public metadata e.g. usernames and the average content posted on social platforms along with the content within a specified timeframe. The system standard format confirms the metadata has gone through the system in a processed formation.

4. Rule-Based Analysis: The normalized data is further processed using a core rule-based analytical engine which compares this profile against a set of heuristics obtained from research studies involving actual social media behavior. Some of the elements investigated include profile completeness, follower-following ratio, username entropy, content similarity, and post frequency. Each heuristic augments the profile evaluation and, when summarized, produce a risk index that estimates the probability of the account being a fraud. The design of the system supports explainable AI such that every classification decision is clear and the reasoning traceable.

5. Result Storage: A combination of automated system results, timestamps, and metadata are stored using the PostgreSQL database via Django's Object Relational Mapper (ORM). This system aims to preserve transactional consistency and optimize the system when the data is to be extracted for review, performance evaluation, auditing, or heuristic changes. On top of this focus concerning storage of data, this method of storage supports a time series of events for temporal analytics, detection of patterns that blossom over.

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6. Administrative Review: Review of Reports. Due to the auto classification of the reports, administrators know of the classification assigned to the report prior to viewing the report. Therefore, administrators can log in, to the secure dashboard, and compare the analytical reports to the reports, to approve, adjust, comment on the auto classification. The blended system of automation and administration distinctively removes the concern of possible false positive automation. It also adds a positive balance to the equality aspect of the report classification. Also, the administrative classification adds identifiable governance and transparency of the system.

7. Notification to User: Review of Reports. The system generates a short summary of the analytical review, includes the assessed indicators, the system's final classification and the associated possible concerns, and returns the summary to the report originator. This summary promotes transparency and awareness of a positive digital safety. This summary also closes the participatory circuit of allowing users to continue the review process. This is particularly helpful to users reporting suspected fraudulent social media accounts.

VI. SYSTEM IMPLEMENTATION

In building the financial model of the technology stack, the design was purposefully the result of an integration of open-source technology and a balance of cost, flexibility, and maintainability. Layer by layer, the technology stack was articulated to best balance trade-offs among performance, compatibility, and accessibility.

1. Backend: For the secondary development language, I chose Python, due to its modular design and availability of strong libraries. Using Django to build the backend, I implemented an MVT (Model View Template) architectural pattern which allows for secured, structured development and expedited deployment.

2. Frontend: Using HTML5, CSS 3, and JavaScript and associated web development tools, I will build a User Interface. Its design is entirely mate and does not interfere in any way with how well the design adapts to the web's agnosticism and responsive design in regard to screen size. The platform is open to users of all levels of tech skills.

3. Database: Our DBMS makes use of a PostgreSQL which is a DBMS known for its impressive public domain and query optimization and its ACID commitment to data integrity. It retains analytical outputs, user data, and system audit data with transaction level consistency under Django's Object Relational Mapper (ORM) and encrypted data.

VII. RESULTS AND ANALYSIS

System tests prove its efficacy in the recognition of fraudulent social media profiles and the indicators listed demonstrate commendable results in the basic measures of fraud detection. One of the strengths of the model is the accessibility of the reasoning behind the outcomes. Users and admins are able to seamlessly trace the logic from the raw profile information to the risk score. This clarity contrasts with the black-box nature of the reasoning behind the outcomes of intricate neural nets. The model showed reliability in detecting the features of accounts that disseminated spam and engaged in synchronized bot activity and heuristically flagged accounts that were behaving and structured anomalously.

The results of the computer simulations indicate that the model's design is indeed purposeful, sparing an amount of memory that is dwarfed by that of deep learning approaches, and in relation to the level of moderate memory demands the model is computationally efficient. This is a design objective meant to enable the framework to remain both open and extensible in terms of minimal hardware requirements.

Table 1 : Comparative Analysis with Existing Detection Algorithms

Model / Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Remarks
Proposed Rule-Based Model	98.8	99.5	97.2	98.8	High transparency; low computation cost
CNN–LSTM Hybrid Model [2]	95.4	94.1	92.7	93.4	High accuracy; limited interpretability
Transformer-Based Model [8]	96.2	95.6	94.3	94.9	Handles dynamic behaviors efficiently
GNN-Based Detection [3]	94.8	93.2	92.1	92.6	Good relational behavior analysis

The rule-based framework effectiveness was quantitatively assessed relying on a labeled set of 5,000 social media profiles. The following are the metrics we calculated to evaluate classification by the system; all of them were based on those in the confusion matrix:

A. Evaluation Metrics

In order to analyze the performance achieved with the system, we applied the following formulas: Accuracy is evaluation of overall correct classifier and defined as: **Accuracy** is evaluation of overall correct classifier and defined as:

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$$\textbf{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%$$

Precision is a measure of the overall quality (in this case, accuracy and likelihood of localization) of our positive (fake) predictions:

$$\textbf{Precision} = \frac{TP}{TP + FP} \times 100\%$$

Recall (Sensitivity): Recall is the measure of the ability to detect all the positive instances:

$$\textbf{Recall} = \frac{TP}{TP + FN} \times 100\%$$

1. **F1-Score** is the harmonic mean of Precision and Recall:

$$\textbf{F1-Score} = \frac{2 \times (\textbf{Precision} \times \textbf{Recall})}{(\textbf{Precision} + \textbf{Recall})}$$

where:

TP = True Positives (fake profiles being rightfully detected)

TN = True Negatives (Real profile correctly identified)

FP =False Positives (Authentic accounts wrongfully classified as fake)

FN = False Negatives (Fake Profiles Missed by the system)

B. Experimental Results

The system's performance was solid and is characterized in Table 1. High precision also indicates lower false positives which is advantageous for user confidence and trust. Positive magnitude of F1-Score confirms model reliability.

Table.2 Performance Metrics of Training Dataset

Metric	Performance (%)
Accuracy	98.8
Precision	99.5
Recall	97.2
F1-Score	98.8

Examining these metrics closely tells us the about the inner workings of the model. A notable benchmark of 98.8% model accuracy reaffirms that the model ability will likely be able to generalize on profiles that are in the deployment test set. Furthermore, precision was also above average, 99.5% which is a good indicator that false positives will not be a problem. This is a key feature for user trust because false positives lead to user frustration, retention problems. In user retention, false positives are a point of concern as it can lead to reputation loss as some users were flagged as fraudulent when they were legitimate.

The data we collected during the evaluation allow us to firmly ascertain that the clear and rule-based methods satisfy the objectives of achieving both high accuracy in detection while maintaining overall efficiency. The straightforward procedural logic embedded within the heuristic rules allows for the system to analyze profiles quickly and operate at the appropriate speed required to scale and generate real-time system feedback. The performance and efficiency of a system in a transparent manner offers system administrators operational support and allows them to trust the system, while also engaging them with the end-users. This system consequently closes the gap between detection models in theory and a proposed determining system users will trust and engage with.

VIII. TECHNICAL IMPLEMENTAION

This system has been made to be very easy to set up on a standard tech stack, making it very easy to install and run in any sort of setting. The backend has been developed in the Django Framework meaning it can be deployed anywhere a Python virtual environment is set up. With regards to the database management, we propose using PostgresSQL in production as it is stable, scalable, and ACID compliant. On the other hand, we suggest using SQLite for lightweight use cases, for instance, testing or during the development phase. The platform has been integrated with the Twitter and Facebook APIs for real-time data aggregation and data analysis, and the integration of other social media platforms can be used in case further development is needed. In production, we use the Gunicorn WSGI server, and we serve the static files using Nginx in order to reduce the number of requests to the server and to improve its security. With regards to the platform itself, it can be run fairly lightly with about 1GB of Ram, about 1 CPU core, and at least 10GB of disk space, so we recommend around 2GB of Ram and 2 CPU cores for the best experience. This is the sort of low overhead on the infrastructure of the system that makes it a very scalable system, and ultimately very useful in the real world.

IX. FUTURE WORK

Developing this framework guarantees that even more research in this field will be conducted. This allows you to access what you need out there as well. The expected future improvements involve algo incorporating rules with machine learning. Such a hybrid solution would address adaptive detection. The other one is developing a dashboard so that users can readily track patterns associated with fake accounts. And this

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leads to the concept of a browser extension that enables users to instantly report fake accounts directly from their social media feeds. Other improvements include the broadening of the analysis across multiple social media platforms, as well as the detection of fake accounts clustered in the same network. All these changes would greatly enhance the system's functionality and efficiency, possibly even making it self-learning and adaptive to be implemented in any country across the globe.

X. CONCLUSION

This research describes the effective ways of conducting fake profile detection at different levels of tolerable risks. The approach is straightforward and reachable. The verifiability of the explainable process is still within reasonable bounds. The authors constructed and highly academically engaged the audience through the multi-modal rule-based Framework and its heavy components. While devising a risk management framework, they also embedded an analytical component and provided a web-based tool and community-based admin features for their analytical tool for fake profile detection, and the appropriate risk mitigation framework. The authors propose a maintenance-friendly alternative to the technology that is predominated by proprietary 'black boxes'. The authors postulate adequately that good cybersecurity comes with reasonable, logical, and easy to understand designs. With transparency. With inclusion of the end-users. Not with opaque algorithms. Projects like this shape the social media defenses that trust and accessibility. This also takes us a step closer to a digital ecosystem we can trust.

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